AI/ML Internship – Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Metric Selection Justification

For customer churn prediction, the primary business objective is to identify customers who are likely to leave the company. Therefore, **Recall** is considered one of the most important metrics because it measures the percentage of actual churn customers correctly identified by the model.

While Accuracy provides an overall measure of correctness, it can be misleading in imbalanced datasets. Therefore, Precision, Recall, F1 Score, and AUC were also used to evaluate model performance comprehensively.

- **Accuracy** measures overall prediction correctness.
- **Precision** measures how many predicted churn customers actually churned.
- **Recall** measures how many actual churn customers were correctly detected.
- **F1 Score** balances Precision and Recall.
- **AUC (Area Under ROC Curve)** measures the model's ability to distinguish between churn and non-churn customers.

Imbalance Handling

The Telco Customer Churn dataset contains a higher number of non-churn customers than churn customers, creating a class imbalance problem.

To address this issue, a **Balanced Logistic Regression** model was implemented using:

class_weight='balanced'

This technique assigns higher importance to the minority class (churn customers) during training.

The impact can be observed in the results:

- Recall increased from **56.68%** to **78.07%**
- F1 Score improved from **60.92%** to **61.34%**

This demonstrates that the balanced model was more effective at identifying customers who were likely to churn.

Final Model Decision

Three classification models were evaluated for customer churn prediction:

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	80.70%	65.84%	56.68%	60.92%	0.8416
Balanced Logistic Regression	73.88%	50.52%	78.07%	61.34%	0.8412
Decision Tree Classifier	79.42%	62.96%	54.55%	58.45%	0.8284

Although Logistic Regression achieved the highest Accuracy (80.70%) and Precision (65.84%), the Balanced Logistic Regression model achieved:

- Highest Recall (78.07%)
- Highest F1 Score (61.34%)
- Comparable AUC (0.8412)

In a real-world churn prediction system, failing to identify a customer who may leave is generally more costly than incorrectly flagging a customer as a potential churn risk. Therefore, maximizing Recall is more important than maximizing Accuracy alone.

The Balanced Logistic Regression model successfully identified a significantly larger proportion of churn customers while maintaining competitive overall performance.

Final Decision

Balanced Logistic Regression was selected as the final model because it achieved the highest Recall (78.07%) and highest F1 Score (61.34%), making it the most effective model for identifying potential churn customers. Its AUC score (0.8412) was also very close to the standard Logistic Regression model, indicating strong classification capability.

This model provides the best balance between business usefulness and predictive performance for customer churn prediction.